



Produced by
Psion Teklogix Ltd

Calc User Guide

Contents

Introduction	1
General calculations	1
Using the memory	2
Percentage calculations	2
Scientific calculations	2
Operator precedence	3
Re-using answers and calculations	3
Using the memories	3
Powers, roots & reciprocals	4
Trigonometric functions	4
Angle formats	4
Factorials	5
Logarithms	5
Displaying numbers	5
Exponent notation	5

Introduction

Calc is the calculator program. It is split into two views: Desk and Scientific. The two calculators are distinct, so you can carry out two calculations at the same time without them affecting each other.

- The **Desk view** is a desktop calculator which produces a “till roll” output. Use this calculator for general arithmetical calculations, and those involving percentages.
- The **Scientific view** is a scientific calculator with 26 memories. Use this for more advanced calculations, including those that involve trigonometry or logarithms.

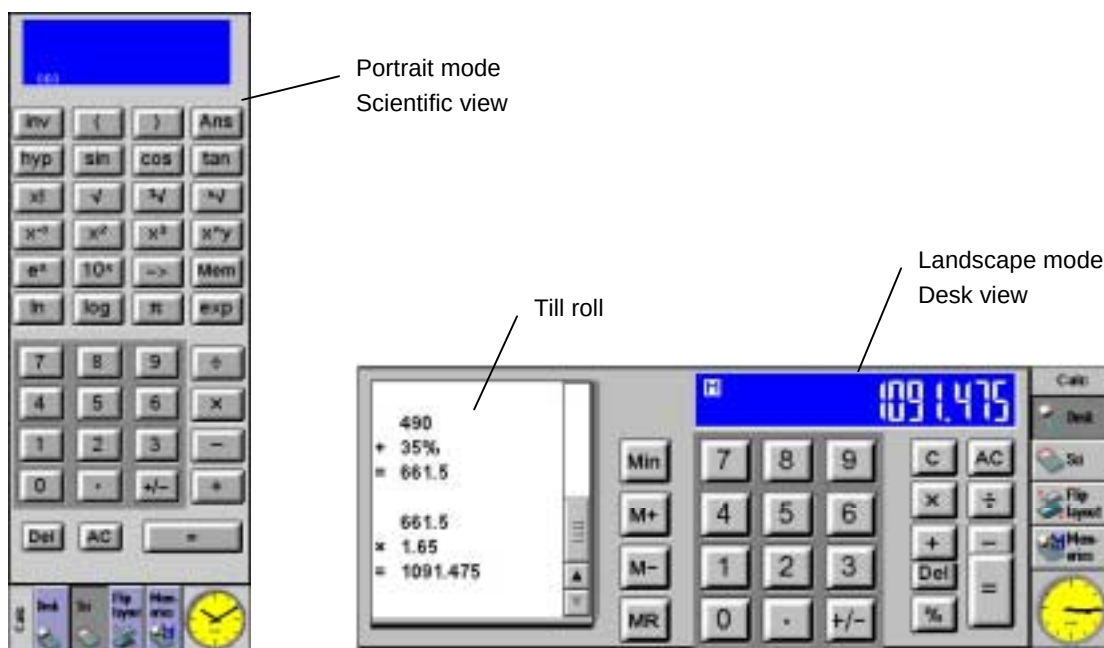
When you first use Calc, it displays the Desk calculator. Use the [Desk](#) and [Sci](#) buttons to move between the views, or select the view you want from the [View](#) menu.

Note: You may find the display suits your style of working better when it is “flipped”, i.e. when the left and right sides of the screen are swapped over. Just tap [Flip layout](#) on the Toolbar, or use the command on the [View](#) menu. If [Flip layout](#) is tapped a second time, the calculator is displayed in portrait mode.



Displaying on-line help

To display on-line help, select the [Help on Calc](#) option from the [Tools](#) menu.



General calculations

To perform a calculation in the Desk calculator:

1. Enter your calculation using the number and operator buttons (+, -, × and ÷) on screen or the keyboard. There is no operator precedence in the Desk view: all calculations are performed as you type them in. Therefore, 10+5×3=45, not 25.
2. Tap = or press Enter to get the result.
 - Use +/- or the m key on Psiboard to change the sign of a number.
 - To clear the calculation line without ending the current calculation, press C.
 - To clear the calculation line and end the current calculation, tap AC or press Esc.

The till roll keeps a record of your calculations so that you can refer back to work you did earlier. The till roll scrollbar will appear when the calculations cover more than one page. To move around the till roll, use the scrollbar or the arrow keys. To clear the till roll, select the **Clear till roll** command on the **Tools** menu.

Using the memory

You can use the memory to store a value that you want to re-use, or to act as a “running total”. The **Memory** commands can be accessed from the **Memories** button on the Toolbar or the command on the **Tools** menu.

- **To store a number in the memory:** tap **Min** or select the **Save in** command while the number you want to store is in the calculation line. An 'M' will appear in the calculator display while there is a value stored in the memory.
- **To retrieve a number:** tap **MR** or select the **Recall** command.
- **To add the current number to the number in the memory:** tap **M+** or select the **Add to** command.
- **To subtract the current number from the number in the memory:** tap **M-** or select the **Subtract from** command.
- **To clear the memory:** tap **Min** while 0 is displayed in the calculation line, or select the **Clear** command.

Percentage calculations

Use **%** on the Desk calculator to perform percentage calculations.

- **To calculate 40% of 60:** enter 60×40 , then tap **%**.
- **To increase 60 by 40%:** enter $60 + 40$, then tap **%**.
- **To decrease 60 by 40%:** enter $60 - 40$, then tap **%**.
- **To calculate what % 60 is of 200:** enter $60 \div 200$, then tap **%**.
- **To find the number that 60 is 40% of:** enter $60 \div 40$, and then tap **%**.

You can use a combination of the memory and the **%** facility for, say, tax calculations. If you have the rate of taxation stored in the memory, you can quickly add or deduct the tax from any number:

- To add the tax to a number: enter the number and tap **+**. Tap **MR** and then **%**.
- To deduct the tax from a number: repeat the above, using **-** instead of **+**.

Scientific calculations

To perform a calculation in the Scientific view:

1. Use the number, operator and function buttons to enter your calculation. Enter scientific functions in the order you would write them down. For example, tap **log** BEFORE entering the number you want to find the log of, but tap **x²** AFTER entering the number you want to square.



2. Tap **=** or press Enter to get the result.
 - **To clear the calculator display:** tap **AC** or press Esc.
 - **To remove the item immediately to the left of the cursor:** press Del.

You can position the cursor anywhere on the calculation line using the pen.

Operator precedence

Operator precedence in the Scientific view is as follows (highest first):

- Information in brackets.
- Functions which follow the value to which they refer, e.g. $x!$
- Powers.
- Implicit multiplication before a memory value, i.e. $30A$.
- Prefix functions that precede the value they refer to, e.g. $\sin$.
- Implicit multiplication before prefix function, such as $5\sin 30$, or before an open bracket, as in $4(4+5)$.
- Multiplication and division equal, calculated left to right.
- Addition and subtraction equal, calculated left to right (lowest).

See the Calc on-line help for more about entering expressions and operator precedence in the Scientific calculator.

Re-using answers and calculations

In the Scientific view you can re-use the answer from your last calculation.

- **To insert the last answer at any point in a calculation:** tap the [Ans](#) button at the appropriate point. The calculator will insert 'Ans' in the calculation line, representing the last result.
- **To start a new expression with the last answer:** just enter the rest of the expression as though the answer was already written at the start. The calculator will prefix the expression with 'Ans'.

You can also re-use and edit calculations you have performed earlier. This can save time if you want to perform a new calculation that differs only slightly from a previous one.

- **To re-use a previous expression:** use the up and down keys, or the [Edit previous](#) and [Edit next](#) commands on the [Edit](#) menu, to display the last 10 calculations one by one on the calculation line. When you find the expression you want to re-use, change it if required and proceed as normal.

Note: The value of Ans used in a calculation will always be the value of the last calculation. If you re-use a calculation based on a value of Ans, the result will be calculated using the current value of Ans rather than the value at the time of the original calculation.

Using the memories

The Scientific calculator has 26 memories, labelled A through to Z. You can assign a value to each of these memories, and then incorporate the names into expressions.

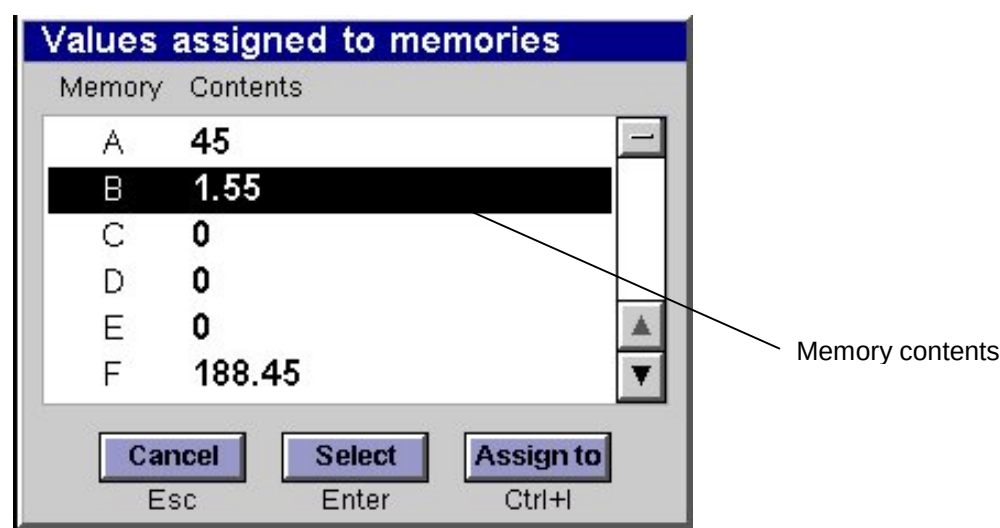
To assign a value to a memory:

1. Enter the number to be stored. If the calculation line is displaying an unsolved expression, the stored value will be the RESULT of the expression.
2. Tap [->](#), or select [Assign to](#) from the [Memories](#) button on the Toolbar or the command on the [Tools](#) menu.
3. Type the memory name. Because memory names must be entered as a capital letter, tap the Shift key and then type the letter key for the memory name.

Note: You can also assign a value to a memory by tapping the [Mem](#) button, highlighting the memory to use and then tapping the [Assign to](#) button in the dialog.

- **To display the contents of the memories:** select [Show all](#) from the [Mem](#) button on the Toolbar, or select [Show all](#) from the [Memories](#) commands on the [Tools](#) menu.
- **To use a stored number in a calculation:** type the capital letter corresponding to the memory at the appropriate point in the calculation. Alternatively, tap the [Mem](#) button to display the list of memories, highlight the memory you want and tap [Select](#).

Note: You can assign values to the memories in any order you like, so use the names as a reminder of the information they contain. For example, use the T memory for a rate of taxation, Y for the rate of exchange into yen and so on.



Powers, roots & reciprocals

You can calculate squares, cubes and powers using the buttons in the Scientific view.

- **To calculate the square of a number:** enter the number, tap x^2 , then press $=$.
- **To calculate the cube of a number:** enter the number, tap x^3 , then press $=$.
- **To calculate a number to a given power:** enter the number you want to raise to a given power. Tap x^y and enter the power. Press $=$.
- **To calculate the square root of a number:** tap $\sqrt{}$, enter the number, then press $=$.
- **To calculate the cube root of a number:** tap $\sqrt[3]{}$, enter the number, then press $=$.
- **To find the x root of a number:** enter the number of the root you want to find, e.g. 4 for the fourth root. Tap $\sqrt[x]{}$, and enter the number you want to find the x root of. Press $=$.
- **To calculate the reciprocal (1/x) of a number:** enter the number then tap x^{-1} and tap $=$.

Trigonometric functions

- **To calculate the sine, cosine or tangent of a number:** tap **sin**, **cos** or **tan**. Enter the number and press $=$.
- **To calculate the inverse sine, cosine or tangent of a number:** tap **inv** once, then tap **sin**, **cos** or **tan**. Enter the number and press $=$.
- **To calculate the hyperbolic sine, cosine or tangent of a number:** tap **hyp** once, then tap **sin**, **cos** or **tan**. Enter the number and press $=$.
- **To calculate the inverse hyperbolic sine, cosine or tangent of a number:** tap **hyp** and **inv** once each to depress them, then tap **sin**, **cos** or **tan**. Enter the number and press $=$.

Angle formats

The scientific calculator can express angles in degrees, radians and gradients (360 degrees = 2π radians = 400 gradients). The angle units you use will affect the outcome of trigonometric calculations.

- **To change the angle format:** select the **Formats** command from the **Tools** menu. Select the format you require.

You can also change the angle format by tapping on the abbreviation in the calculation line (e.g. DEG). If the result of a trigonometric calculation is being displayed when you do this, the result will automatically be re-evaluated.

Factorials

The **x!** button in the Scientific view calculates the factorial of a number. A factorial is the result of multiplying all the numbers from a given starting number down to one. So, the factorial of 4 (written as 4!) is $4 \times 3 \times 2 \times 1 = 24$.

Note: You can use factorials to find the number of ways things can be arranged. For example, if you want to find possible anagrams of a word with four letters, 4! shows that there are 24 ways these letters can be arranged.

- To find the factorial of a number: enter the number of which you want to find the factorial. Tap **x!** and tap **=**.

Logarithms

You can use the Scientific calculator to calculate logs and natural logs.

- **To calculate the log (base 10) of a number:** tap **log**, then enter the number. Press **=**.
- **To calculate the natural log (base e) of a number:** tap **ln**, then enter the number. Press **=**.
- **To calculate e to a given power:** tap **e^x**, then enter the value you want to find e to the power of. Press **=**.

Displaying numbers

The Scientific calculator can display numbers in the following formats:

- **Normal:** this is adequate for most calculations. Numbers are displayed in standard notation to 12 significant figures, with decimal places and an exponent as required.
- **Fixed:** this format always displays results with a specified number of decimal places, e.g. for currency calculations.
- **Scientific:** you may find this useful for dealing with very large or very small numbers. Numbers are displayed as a single digit to the left of the decimal point, a specified number of significant figures, and then an exponent.

To change the number format: select the command from the **Tools** menu. If you choose Fixed or Scientific, you can set the number of decimal places or significant figures to be displayed.

Exponent notation

You can express very large and very small numbers more succinctly using positive and negative exponents. For example, 57 200 000 can be expressed as 5.72E7.

To express a number as an exponent:

1. Enter the first part of the number, normally as a decimal fraction.
2. Tap **exp**. An 'E' will appear on the calculation line. To express a very small number, tap **+/-** to insert a minus sign before the second number.
3. Enter the number of decimal places that the first number needs to be moved to the left.